



Bahir Dar University
Bahir Dar Institute of Technology
Faculty of Computing
Information Technology

Internship Report

Qelem Meda Technologies PLC
BetLink Real Estate & Rental Property Management System

By: **ID**

1. Adisu Dereje..... BDU1509909

Mentor: Mr. Asnakew Lingerih

Company Supervisor: Mr. Samuel Temesgen

Date: 14/01/2019 E.C / September 2026

Declaration

I, the undersigned, declare that this internship report is my own original work, prepared and submitted as part of fulfilling the requirements of our internship program at Qelem Meda Technologies PLC. This report has been written based on the knowledge, skills, and practical experiences I gained during the internship period.

The results, discussions, and conclusions presented in this document reflect my own effort, observations, and understanding of the tasks and projects I carried out during the internship. I have tried to present the information in a clear and honest way to show what I have learned and achieved.

Finally, I take full responsibility for the originality, quality, and correctness of the content in this report. If there are any mistakes, missing points, or errors, I accept accountability for them.

Names and Signatures

1. Adisu Dereje

Signature:



Date: 14/01/2019 E.C

Approved By

1. Academic Mentor: Mr. Asnakew Lingerih

Signature:



Date: 14/01/2019 E.C

Acknowledgements

First and foremost, I offer my supreme praise, honor, and heartfelt thanksgiving to the Almighty God, who granted me life, wisdom, good health, and endurance to start, pursue, and successfully complete this internship period and finalize this comprehensive report. His divine guidance and blessings have been my continuous source of strength throughout this journey.

I would like to express my sincere gratitude to Mr. Samuel Temesgen, my internship supervisor at Qelem Meda Technologies PLC, for his invaluable guidance, continuous support, and expert advice throughout our internship period. His dedication to mentoring and willingness to share his professional knowledge greatly contributed to my understanding of practical work in the field of software engineering. Mr. Samuel patiently supervised my work, provided constructive feedback on my tasks, and encouraged me to explore innovative solutions, which enhanced my learning experience and helped me grow both professionally and personally.

I am also deeply thankful to the entire staff of Qelem Meda Technologies PLC, their cooperation, encouragement, and for creating a friendly and motivating work environment. Their willingness to assist whenever I faced challenges allowed me to gain hands-on experience and practical skills that I would not have acquired otherwise. I greatly appreciate the teamwork and professional attitude exhibited by all staff members, which inspired me to improve my own communication, collaboration, and problem-solving skills.

Furthermore, I extend my heartfelt gratitude to Bahir Dar University, Bahir Dar Institute of Technology, Faculty of Computing, and specifically the Department of Information Technology. Special thanks go to my academic mentor, Mr. Asnakew Lingerih, for his continuous encouragement, constructive guidance, and valuable advice throughout the internship period.

Finally, I would like to extend my heartfelt gratitude to my family and friends, whose understanding, prayers, and motivation were a constant source of strength throughout my academic journey and internship experience. Their encouragement gave me the confidence to face new challenges and persevere in completing my internship successfully.

Overall, this internship experience would not have been as rewarding without the support and guidance of my supervisor, my academic mentor, the staff of Qelem Meda, and my loved ones. I am sincerely grateful to all who contributed, directly or indirectly, to my learning and professional growth during this period.

Executive Summary

This internship report presents the collective experiences, practical skills, and learning outcomes I gained during my internship at Qelem Meda Technologies PLC, an emerging and ISO-certified software development company in Ethiopia. The main objective of the internship was to acquire practical knowledge and hands-on experience in the field of information technology, particularly in full-stack web application development, database management, and modern software engineering practices, while applying theoretical concepts learned during my academic studies to real-world scenarios.

During my internship, I was assigned to the System Development Department within the Web Application Development Unit, where I actively participated in various tasks including database modeling, backend RESTful API development using Laravel 13, and modern Single Page Application (SPA) frontend development using Vue.js 3 and Vite. Under the guidance of Mr. Samuel Temesgen, my company supervisor, I gained a strong understanding of modern software engineering practices, clean code architecture, and the process of creating scalable web applications. My primary project engagement was working on the BetLink platform — a comprehensive web-based Real Estate and Rental Property Management System developed to address inefficiencies, lack of transparency, and informal brokerage in property transactions across urban Ethiopia.

Throughout the internship, I faced several technical challenges such as handling concurrent appointment scheduling conflicts, ensuring cross-browser responsiveness, integrating two-factor OTP verification, and optimizing high-resolution property media assets for bandwidth-constrained environments. Overcoming these challenges required critical thinking, disciplined debugging, pessimistic database locking, and continuous collaboration with senior developers.

This experience also helped me strengthen essential professional skills. I improved my communication and teamwork abilities through daily standups and sprint reviews, enhanced my technical expertise in Laravel and Vue.js, developed leadership qualities by taking ownership of assigned feature modules, and gained a deeper appreciation for work ethics, software security, and client-focused service delivery. In conclusion, this internship provided me with hands-on, practical engineering experience that connected what I learned at Bahir Dar Institute of Technology with the day-to-day realities of software development in the industry, preparing me to enter the ICT sector with practical competence, technical confidence, and professional discipline.

Contents

Declaration	i
Acknowledgements	ii
Executive Summary	iii
List of Tables and Figures	vi
Chapter One: Company Background of the Internship Hosting Organization.....	1
2.1 Introduction.....	1
2.2 Brief History of Qelem Meda Technologies PLC	1
2.3 Vision.....	2
2.4 Mission.....	2
2.5 Core Values	2
2.6 Main Products and Services	3
2.7 Main Customers and End Users	6
2.8 Organizational Structure	7
2.9 Functional Units and Their Responsibilities.....	10
2.10 Workflow of Functional Units	12
2.11 Technologies Used by the Company	16
2.12 Chapter Summary.....	16
Chapter Two: Internship Experience.....	18
2.1 Introduction.....	18
2.2 Placement Section and Reasons.....	18
2.3 Work Tasks Executed.....	19
2.4 Engineering Methods, Tools, and Techniques Used	20
2.5 Major Challenges and Problems Faced	20
2.6 Measures Taken and Recommendations.....	22
2.7 Personal Gains and Development	24
2.8 Recommendations and Conclusion	25
Chapter Three: Project Work – BetLink Property Management System	26
3.1 Summary of the Project.....	26
3.2 Problem Statement and Justification	27
3.3 Objectives of the Project	28
3.4 Methodology	29
3.5 Analysis, Results, and Discussion	30
3.6 Proposed Solution	31
Chapter Four: General Conclusion and Recommendation	32
4.1 General Conclusion	32

4.2 General Recommendations.....	32
4.3 Final Remarks	33
References.....	34
Appendices	35
Appendix A: Sample Screenshots of the BetLink Platform	35
Appendix B: Code Snippets and Features Implemented	35
Appendix C: Flowcharts and System Architecture Diagrams	37

List of Tables and Figures

Tables

Table 1.1: Summary of Key Services Offered by Qelem Meda Technologies PLC	6
Table 1.2: Objectives of the BetLink Property Management System.....	28

Figures

Figure 1.1: Organizational Structure Hierarchy of Qelem Meda Technologies PLC	8
Figure 1.2: Executive Leadership Team of Qelem Meda Technologies PLC.....	9
Figure 1.3: SDLC Workflow of Qelem Meda Technologies PLC.....	14
Figure 1.4: Qelem Meda Technologies Engineering Team Review.....	16
Figure 2.1: The author (Adisu Dereje) conducting web engineering tasks at Qelem Meda Technologies PLC.....	19
Figure 2.2: Quality assurance testing and error validation on Haleta ERP with Notion defect tracking (Adisu Dereje).....	24
Figure 3.1: The BetLink Real Estate and Rental Property Management System Hero Interface.....	26
Figure 3.2: BetLink Verified Property Catalog and Filtering Interface	27
Figure 3.3: BetLink Buyer & Tenant Dashboard.....	30
Figure 3.4: Data Flow Diagram for Tour Appointment Scheduling.....	31

Chapter One: Company Background of the Internship Hosting Organization

2.1 Introduction

This chapter provides an overview of Qelem Meda Technologies PLC (QMT), the organization where I completed my internship. It describes the company's history, mission, vision, organizational structure, products and services, customer base, operational workflow, and the technologies it employs. Understanding the company's background provides valuable context for the internship experience and demonstrates how modern software engineering principles are applied in a professional software development environment.

2.2 Brief History of Qelem Meda Technologies PLC

Qelem Meda Technologies PLC (QMT) is a privately owned Ethiopian technology company established in 2012 Ethiopian Calendar (E.C.) with the objective of delivering innovative software solutions that simplify complex organizational processes and accelerate digital transformation. The company was founded by professionals committed to using technology to solve practical challenges faced by government institutions, educational organizations, healthcare providers, transportation agencies, and private enterprises.

Since its establishment, QMT has grown into one of Ethiopia's emerging software companies by developing secure, scalable, and high-performance software systems tailored to clients' specific requirements. The company focuses on delivering enterprise-level applications that improve organizational efficiency, automate manual processes, enhance decision-making, and promote digital transformation.

Over the years, QMT has successfully delivered more than thirty software solutions, many of which are actively used by institutions across Ethiopia. The company has earned recognition for producing reliable software systems that comply with international quality and information security standards. Its commitment to excellence is demonstrated by achieving ISO/IEC 27001:2022 certification for Information

Security Management Systems and ISO 9001:2015 certification for Quality Management Systems. These certifications reflect the company's dedication to delivering secure, reliable, and high-quality software products and services.

One of the company's flagship products is the Exam Management and Correction System (EMaCS), a comprehensive platform that digitalizes examination management by supporting question preparation, candidate registration, examination administration, automated correction, result publication, and certificate generation. The system has been deployed across multiple educational institutions and regional education bureaus, significantly improving the efficiency and transparency of examination processes.

Today, Qelem Meda Technologies continues to expand its portfolio by developing enterprise software, education management platforms, ERP systems, mobile applications, cloud-based solutions, and customized software products that contribute to Ethiopia's digital transformation agenda.

2.3 Vision

The vision of Qelem Meda Technologies PLC is:

"To be one of the leading technology companies in making clients' lives easier."

This vision reflects the company's commitment to leveraging modern technology to simplify organizational processes, improve operational efficiency, and deliver innovative solutions that create value for its customers.

2.4 Mission

The mission of Qelem Meda Technologies PLC is:

"To be one of the leading technology companies in making clients' lives easier."

The company fulfills this mission by designing and delivering secure, scalable, user-friendly, and reliable software solutions that address real-world challenges while maintaining high standards of quality, security, and customer satisfaction.

2.5 Core Values

Qelem Meda Technologies operates according to a set of core values that guide its organizational culture and software development practices. These values include:

- • **Customer Focus** – Understanding client requirements and delivering solutions that create measurable value.
- • **Problem Solving** – Developing innovative solutions for complex real-world challenges.
- • **Trust** – Building long-term relationships through transparency, integrity, and reliability.
- • **Teamwork** – Encouraging collaboration among multidisciplinary teams to achieve common goals.
- • **Commitment** – Maintaining dedication to quality, professionalism, and timely delivery.
- • **Innovation** – Continuously embracing new technologies, methodologies, and creative ideas.
- • **Respect** – Promoting mutual respect among employees, clients, and stakeholders.

The company's motto,

"Let's Make Life Easier,"

reflects its commitment to improving people's lives through practical technology solutions.

2.6 Main Products and Services

Qelem Meda Technologies PLC (QMT) is a software development company that specializes in designing, developing, implementing, and maintaining digital solutions for both public and private sector organizations. The company provides end-to-end software engineering services that help organizations automate business processes, improve service delivery, enhance operational efficiency, and support digital transformation initiatives across Ethiopia.

The company develops customized software solutions based on clients' specific requirements while ensuring scalability, security, usability, and long-term

maintainability. Its multidisciplinary development teams follow modern software engineering practices to deliver high-quality software products that meet international standards.

The major products and services offered by Qelem Meda Technologies PLC include the following:

2.6.1 Enterprise Software Development

QMT develops customized enterprise applications that automate organizational workflows and improve productivity. These systems enable institutions to manage their daily operations efficiently while reducing manual processes and improving decision-making through digital information management.

2.6.2 Education Management Systems

Education is one of the company's primary focus areas. QMT develops comprehensive digital platforms that support educational institutions in managing student information, examinations, admissions, academic records, learning processes, and institutional administration. One of its flagship products is the Exam Management and Correction System (EMaCS), which has been implemented in several regional education bureaus. The system automates examination management activities, including candidate registration, question preparation, examination administration, digital marking, result processing, and certificate generation. This solution has significantly improved the transparency, efficiency, and reliability of examination management.

2.6.3 Health Information Systems

QMT develops digital healthcare solutions that assist hospitals, clinics, and other healthcare institutions in managing patient information, appointments, medical records, reporting, and service delivery. These systems contribute to improving healthcare efficiency and reducing administrative workload.

2.6.4 Government Digital Transformation Solutions

The company collaborates with government organizations by developing software systems that modernize public services through process automation, digital record

management, workflow optimization, and electronic service delivery. These solutions support Ethiopia's ongoing digital transformation initiatives.

2.6.5 Mobile and Web Application Development

QMT designs and develops modern web-based and mobile applications using current software engineering technologies. These applications are designed with responsive user interfaces, secure backend architectures, and scalable infrastructures that enable organizations to serve users efficiently across multiple platforms.

2.6.6 System Integration and API Development

The company provides system integration services that enable different software platforms to communicate and exchange information securely. Through RESTful APIs and integration technologies, QMT ensures seamless interoperability between enterprise systems and third-party applications.

2.6.7 Cloud-Based Solutions

To improve accessibility, scalability, and reliability, QMT develops cloud-enabled software solutions that allow organizations to securely manage and access their information from different locations while maintaining high system availability.

2.6.8 Software Maintenance and Technical Support

In addition to software development, the company offers continuous maintenance, system upgrades, technical support, user training, and performance optimization services. These services ensure that deployed systems remain secure, reliable, and aligned with evolving client requirements.

2.6.9 Information Security and Quality Assurance

QMT places strong emphasis on software quality and information security throughout the software development lifecycle. The company follows internationally recognized standards, including ISO/IEC 27001:2022 for information security management and ISO 9001:2015 for quality management, ensuring that every software solution is secure, reliable, and meets customer expectations.

Overall, Qelem Meda Technologies PLC delivers comprehensive technology solutions that enable organizations to streamline operations, improve efficiency, and accelerate digital transformation. Its broad range of products and services demonstrates the company's commitment to innovation, customer satisfaction, and technological excellence.

Table 1.1: Summary of Key Services Offered by Qelem Meda Technologies PLC

Service	Short Description	Target Customers
Enterprise Software	Customized ERP and process automation workflows to streamline corporate operations.	Government agencies, corporate enterprises, logistics providers.
Education Systems (EMaCS)	Digital exam registration, automated OMR marking, grading, and instant certification.	Regional Education Bureaus, national testing agencies, universities.
Health Information Systems	Electronic medical records, patient appointment queues, and clinical reporting tools.	Hospitals, diagnostic clinics, regional health bureaus.
Government Digital Platforms	E-governance portals, public service workflows, and electronic document repositories.	Municipal offices, regional sector bureaus, public institutions.
Mobile & Web Development	Cross-platform mobile apps (Flutter) and responsive enterprise web portals (Vue/Laravel).	Commercial enterprises, startups, educational institutions.
System Integration & APIs	Secure RESTful API gateways, payment integrations, and microservices bridges.	Financial institutions, telecom operators, enterprise clients.
Cloud & DevOps Services	High-availability cloud architectures, secure hosting, data backups, and disaster recovery.	SMEs, educational institutions, public administrative bodies.
Security & Quality Assurance	ISO 27001 / ISO 9001 compliant code audits, vulnerability assessment, penetration testing.	Banks, public agencies, enterprise platforms, NGOs.

2.7 Main Customers and End Users

The success of Qelem Meda Technologies PLC is reflected in its diverse customer base, which includes government institutions, educational organizations, healthcare facilities, transportation agencies, private companies, and non-governmental organizations. By

serving clients from different sectors, the company has established itself as a trusted technology partner in Ethiopia's digital transformation journey.

The primary customers and end users of the company's products and services include:

- Government institutions, which use QMT's software solutions to automate administrative processes, improve public service delivery, and manage organizational information efficiently.
- Universities and higher education institutions, which utilize digital platforms for research management, student information systems, examination management, and institutional administration.
- Regional Education Bureaus and schools, which implement systems such as the Exam Management and Correction System (EMaCS) to manage examinations, student records, and educational data more effectively.
- Hospitals and healthcare institutions, which use health information systems to manage patient records, appointments, reporting, and healthcare services.
- Private companies and enterprises, which rely on customized enterprise software solutions to improve operational efficiency, workflow automation, and business management.
- Research institutions and development organizations, which benefit from specialized information management systems designed to support research activities, project management, and data-driven decision-making.

The end users of these systems include administrators, managers, employees, researchers, healthcare professionals, educators, students, government officials, and members of the public who interact with digital services developed by the company.

By understanding the operational needs of these diverse stakeholders, Qelem Meda Technologies PLC develops user-centered software solutions that are secure, scalable, and capable of addressing real-world organizational challenges.

2.8 Organizational Structure

The organizational structure of a company defines the hierarchy of authority, communication channels, and responsibilities assigned to different departments. Qelem Meda Technologies PLC adopts a well-structured organizational framework that

promotes effective coordination, efficient decision-making, accountability, and collaboration among its various functional units. This structure enables the company to deliver high-quality software products and services while maintaining international standards of quality and information security. Figure 1.1 illustrates the comprehensive organizational structure hierarchy of the enterprise.

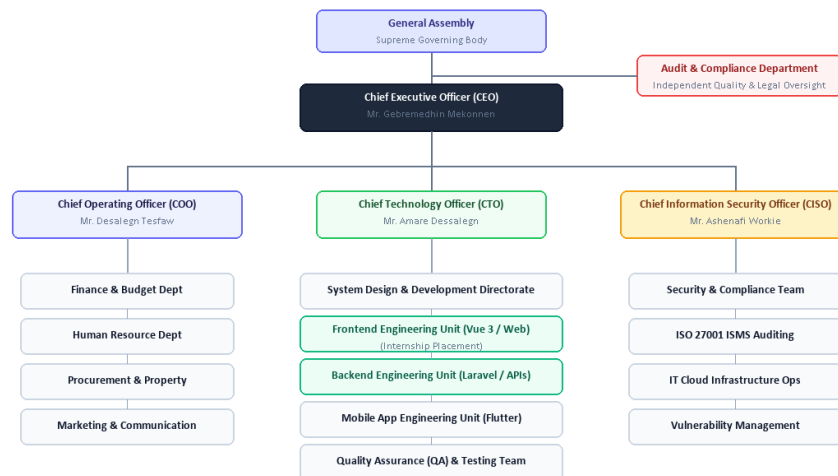


Figure 1.1: Organizational Structure Hierarchy of Qelem Meda Technologies PLC

At the highest level of the organization is the General Assembly, which serves as the supreme governing body responsible for strategic oversight and major organizational decisions. Supporting the General Assembly is the Audit and Compliance Department, which ensures that the company's operations comply with legal requirements, organizational policies, and international quality standards.

The day-to-day operations of the company are led by the Chief Executive Officer (CEO), who is responsible for the overall management of the organization, strategic planning, organizational growth, and business development. The CEO is supported by the Executive Secretary, who coordinates administrative activities and facilitates communication between departments.

To ensure efficient operations and strategic execution, the CEO works closely with the executive leadership team comprising the Chief Operating Officer (COO), Chief Technology Officer (CTO), and Chief Information Security Officer (CISO). As illustrated in Figure 1.2, the executive leadership team features Mr. Gebremedhin Mekonnen (Chief

Executive Officer), Mr. Desalegn Tesfaw (Chief Operating Officer), Mr. Amare Dessalegn (Chief Technology Officer), and Mr. Ashenafi Workie (Chief Information Security Officer). Each executive oversees specialized departments that collectively contribute to the successful execution of the company's objectives.

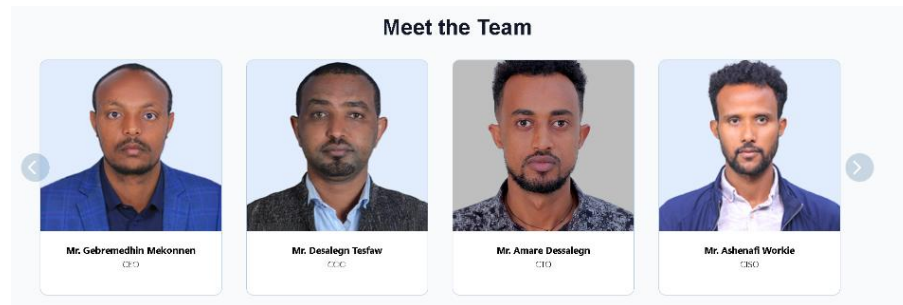


Figure 1.2: Executive Leadership Team of Qelem Meda Technologies PLC (Mr. Gebremedhin Mekonnen - CEO, Mr. Desalegn Tesfaw - COO, Mr. Amare Dessalegn - CTO, Mr. Ashenafi Workie - CISO)

The organizational structure also includes specialized departments such as the Research and Development (R&D) Department, Legal Services Department, Project Management Department, Product Management Department, and the System Design and Development Directorate. These departments collaborate throughout the software development lifecycle, beginning with project initiation and continuing through analysis, design, implementation, testing, deployment, and maintenance.

One of the core operational units of the company is the System Development Department, which consists of specialized development teams responsible for different areas of software engineering. These include the Frontend Development Team, Backend Development Team, and Mobile & Desktop Application Development Team. Working together, these teams design, develop, integrate, and maintain software solutions that meet client requirements and industry best practices.

Quality assurance is maintained by the QA and Testing Team, which performs software verification, validation, and performance testing before systems are released to clients. The IT Operation Team supports infrastructure management, system deployment, server administration, and technical support to ensure stable and reliable system performance.

The company also maintains strong business support departments under the supervision of the Chief Operating Officer (COO). These include the Human Resource Department, Finance and Budget Department, Procurement and Property Department, Marketing and Sales Department, and the Communication Department. Together, these departments ensure efficient resource management, financial control, staff development, procurement processes, marketing strategies, and effective internal and external communication.

To maintain information security and compliance with international standards, the Chief Information Security Officer (CISO) oversees the Security and Compliance Team, which is responsible for implementing cybersecurity policies, protecting organizational information assets, managing security risks, and ensuring compliance with ISO/IEC 27001:2022 standards.

Overall, the organizational structure of Qelem Meda Technologies PLC promotes effective teamwork, clear communication, accountability, and efficient project execution. By clearly defining the responsibilities of each department and encouraging cross-functional collaboration, the company successfully delivers innovative, secure, and high-quality software solutions that satisfy client needs and support Ethiopia's digital transformation initiatives.

2.9 Functional Units and Their Responsibilities

Qelem Meda Technologies PLC is composed of several functional units that work together to achieve the organization's objectives. Each department has clearly defined responsibilities that contribute to the successful delivery of software products and services.

2.9.1 Chief Executive Office (CEO)

The Chief Executive Office provides strategic leadership and overall management of the company. It oversees organizational planning, decision-making, resource allocation, partnership development, and ensures that all departments operate efficiently toward achieving the company's vision and mission.

2.9.2 Chief Operating Office (COO)

The Chief Operating Office manages the company's daily administrative and operational activities. It supervises departments responsible for finance, human resources, procurement, communication, and marketing to ensure smooth organizational operations.

2.9.3 Chief Technology Office (CTO)

The Chief Technology Office is responsible for the company's technological direction. It oversees software architecture, technology selection, software engineering standards, innovation initiatives, and research into emerging technologies.

2.9.4 Project Management Department

This department coordinates software projects from initiation to completion. It prepares project schedules, allocates resources, manages risks, monitors progress, and ensures that projects are delivered within scope, budget, and timeline.

2.9.5 Product Management Department

The Product Management Department gathers customer requirements, prioritizes product features, develops product roadmaps, and ensures that software products meet customer expectations and market demands.

2.9.6 System Design and Development Directorate

This directorate is responsible for designing and implementing software systems. It coordinates software analysis, system architecture, coding, integration, documentation, and deployment activities.

2.9.7 System Development Department

The System Development Department develops software solutions using modern programming languages, frameworks, and development methodologies. It consists of specialized teams including: Frontend Development Team, Backend Development Team, and Mobile & Desktop Application Development Team. Each team focuses on its respective area while collaborating closely to produce complete software solutions.

2.9.8 Quality Assurance and Testing Team

The QA and Testing Team ensures software quality by conducting functional testing, integration testing, performance testing, usability testing, and regression testing before deployment.

2.9.9 IT Operation Team

The IT Operation Team manages servers, deployment environments, networking infrastructure, cloud services, backups, and technical support to ensure continuous system availability.

2.9.10 Research and Development (R&D) Department

The Research and Development Department investigates emerging technologies, explores innovative software solutions, and supports continuous improvement in the company's products and development processes.

2.9.11 Human Resource Department

This department manages employee recruitment, training, performance evaluation, staff welfare, and professional development.

2.9.12 Finance and Budget Department

The Finance Department is responsible for budgeting, financial planning, accounting, payroll management, and financial reporting.

2.9.13 Marketing and Sales Department

This department promotes the company's products and services, builds customer relationships, identifies business opportunities, and supports organizational growth.

2.9.14 Security and Compliance Team

The Security and Compliance Team protects organizational information assets by implementing cybersecurity policies, conducting security assessments, and ensuring compliance with international standards and regulatory requirements.

2.10 Workflow of Functional Units

Qelem Meda Technologies PLC follows a structured and collaborative software development workflow to ensure that every project is delivered efficiently, securely, and according to client requirements. The company applies modern software

engineering principles and encourages continuous communication among departments throughout the Software Development Life Cycle (SDLC). Each department contributes specific expertise while working together toward a common objective of delivering high-quality software solutions. The general workflow of the company is illustrated through the following stages.

2.10.1 Requirement Gathering and Business Analysis

Every software project begins with understanding the client's business needs and objectives. The Project Management Department, together with the Business Analysis (BA) Team and Product Owner, communicates directly with clients to collect functional and non-functional requirements. During this phase, meetings and discussions are conducted to identify business problems, understand user expectations, define project scope, estimate resources, and document system requirements. The collected requirements are then reviewed and approved before moving to the next stage.

2.10.2 System Analysis and Design

Once the project requirements are finalized, the System Design and Development Directorate analyzes the proposed solution and prepares the overall system architecture. This phase involves preparing Software Requirement Specifications (SRS), designing the database schema, creating system architecture diagrams, designing APIs, preparing user interface (UI) wireframes, identifying software modules, and selecting appropriate technologies and frameworks. These design activities ensure that the development team has a clear technical roadmap before implementation begins.

2.10.3 Software Development

After the design phase is approved, the project enters the implementation stage. The System Development Department is responsible for coding and integrating all software components. Development activities are divided among specialized teams: Frontend Development Team develops responsive and user-friendly interfaces; Backend Development Team develops server-side logic, business rules, APIs, authentication, and database interactions; Mobile and Desktop Team develops platform-specific apps. During development, developers regularly collaborate, conduct code reviews, and

integrate their work using version control systems to maintain consistency and code quality.

2.10.4 Quality Assurance and Testing

After software modules are completed, the project is handed over to the Quality Assurance (QA) and Testing Team. The team performs various testing activities, including functional testing, integration testing, system testing, user acceptance testing (UAT), performance testing, and regression testing. Any identified defects are documented and returned to the development teams for correction. This iterative process continues until the software satisfies all quality requirements.

2.10.5 Deployment and System Operation

Once testing is successfully completed, the IT Operation Team prepares the production environment and deploys the software system. Deployment activities include server configuration, database migration, application deployment, security configuration, performance monitoring, backup configuration, and system availability verification. The deployed system is continuously monitored to ensure stable and secure operation.

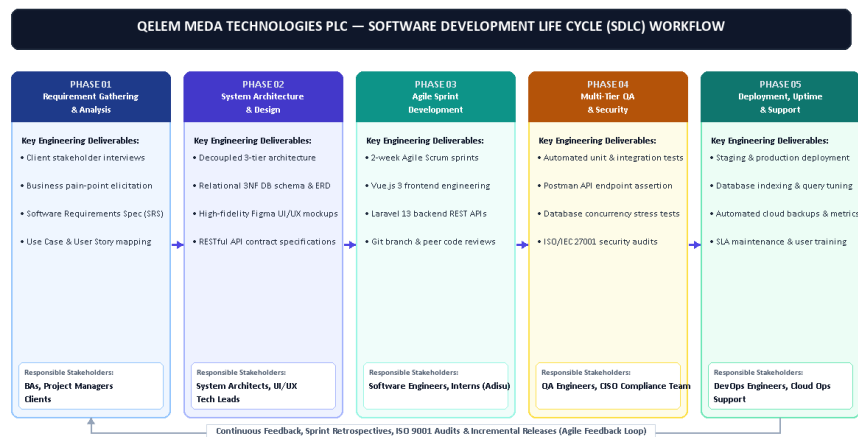


Figure 1.3: SDLC Workflow of Qelem Meda Technologies PLC

2.10.6 Product Management and Customer Support

Following deployment, the Product Management Department works closely with clients to collect feedback, prioritize enhancement requests, and plan future software

improvements. The Product Support Team provides technical assistance, resolves user issues, and performs routine system maintenance to ensure customer satisfaction.

2.10.7 Information Security and Compliance

Throughout every stage of software development, the Security and Compliance Team ensures that all systems comply with organizational security policies and international standards. The team performs security assessments, risk analysis, vulnerability management, access control verification, information security monitoring, and compliance auditing. This continuous oversight helps protect organizational data and ensures the reliability and trustworthiness of the company's software solutions.

2.10.8 Continuous Collaboration

One of the defining characteristics of Qelem Meda Technologies PLC is its collaborative working culture. Rather than operating independently, departments maintain constant communication throughout the project lifecycle. Project managers coordinate activities across teams, software developers collaborate during implementation, quality assurance engineers validate software quality, and management continuously monitors project progress. Regular meetings, progress reviews, code discussions, and technical feedback sessions ensure that all stakeholders remain aligned with project objectives and customer expectations. This collaborative workflow promotes innovation, improves software quality, minimizes development risks, and enables the company to deliver reliable software solutions within the planned schedule.



Figure 1.4: Qelem Meda Technologies Engineering Team Review

2.11 Technologies Used by the Company

Qelem Meda Technologies PLC employs modern software development technologies, frameworks, and engineering tools to design, develop, deploy, and maintain secure and scalable software systems. The selection of these technologies enables the company to build high-performance applications that meet international software engineering standards and satisfy diverse client requirements. The company utilizes a combination of programming languages, frameworks, databases, cloud technologies, development tools, and project management platforms throughout the software development lifecycle:

- **Programming Languages:** PHP, JavaScript, TypeScript, Java, Python, Dart, SQL
- **Frontend Technologies:** Vue.js, React, HTML5, CSS3, Tailwind CSS, Bootstrap, Vite
- **Backend Technologies:** Laravel, Node.js, Express.js, RESTful APIs
- **Mobile Development:** Flutter, Android
- **Database Management Systems:** MySQL, PostgreSQL
- **Version Control and Collaboration:** Git, GitHub
- **Development Tools:** Visual Studio Code, Postman, Composer, npm, Docker (where applicable)

The company follows modern software engineering methodologies and practices, including Agile software development, Scrum-based project management, version control using Git, code review and peer collaboration, continuous testing and debugging, secure software development practices, RESTful API architecture, and database normalization and optimization. These technologies and practices enable Qelem Meda Technologies PLC to develop secure, maintainable, and scalable software solutions while ensuring efficient collaboration among development teams.

2.12 Chapter Summary

This chapter presented the background of Qelem Meda Technologies PLC, including its history, mission, vision, core values, products and services, customer base,

organizational structure, departmental workflow, and technologies used in software development. The chapter demonstrates how the company has established a structured and collaborative environment that supports innovation and the delivery of high-quality digital solutions. Understanding the company's organization and software development process provides a strong foundation for the next chapter, which describes my internship experience and the practical knowledge and skills I gained while working on the BetLink Real Estate and Rental Property Management System.

Chapter Two: Internship Experience

2.1 Introduction

The internship at Qelem Meda Technologies PLC provided me with a valuable opportunity to apply the academic concepts I learned at Bahir Dar Institute of Technology into practical, real-world software engineering. Working in an active company environment allowed me to see how object-oriented programming, database design, web application development, and software architecture come together to build real products.

I was assigned to the System Development Department, focusing on full-stack web application development using Laravel 13 and Vue.js 3. During my internship period, I worked actively on the BetLink Real Estate and Rental Property Management System, participating in requirement discussions, database schema design, RESTful API development, frontend component implementation, testing, and bug fixing. This experience gave me a clear perspective on the complete software development lifecycle in a professional setting.

2.2 Placement Section and Reasons

I was placed in the System Development Department within the Web Application Development Unit. This placement directly aligned with my academic background in Information Technology and my focus on modern web engineering. The department is responsible for architecting and implementing enterprise software solutions and custom client platforms.

Key reasons for my placement included:

- **1. Hands-on Development Experience:** Practical application of modern backend (PHP/Laravel 13), frontend (Vue 3/Vite/Pinia), and relational database (MySQL 8) technologies in an industry-grade environment.
- **2. Exposure to the Product Lifecycle:** Experiencing the end-to-end development cycle from initial requirement analysis and API schema design to code reviews, testing, and deployment.
- **3. Professional Growth and Mentorship:** Learning professional software engineering workflows, Git version control branching standards, and sprint-based project execution from experienced mentors.
- **4. Problem Solving and Innovation:** Tackling real architectural problems such as concurrent appointment booking collisions, database transaction integrity, and responsive frontend state synchronization.

2.3 Work Tasks Executed

During my internship, I executed multiple technical tasks across both backend and frontend layers:

- **A. Database Schema Design & Normalization:** Designed and migrated the MySQL relational schema for properties, units, amenities, user profiles, inspection appointments, and inquiries, enforcing foreign key constraints and composite indexes.
- **B. Backend RESTful API Development:** Implemented secure CRUD controllers in Laravel 13 for listing creation, multi-image uploads with server-side validation, appointment scheduling with transaction locking, and role-based access control.
- **C. Frontend Component Architecture:** Built modular, reusable Vue 3 Single Page Application components including property card grids, interactive filter sidebars, dynamic appointment booking modals, and user management dashboards.
- **D. State Management & API Integration:** Created centralized Pinia stores for authentication state, property catalogs, and booking queues, utilizing Axios interceptors for JWT token injection and centralized error handling.
- **E. Testing, Debugging, and Optimization:** Conducted API functional testing via Postman collections, wrote feature tests for appointment creation endpoints, performed UI cross-browser testing, and resolved responsive layout defects.

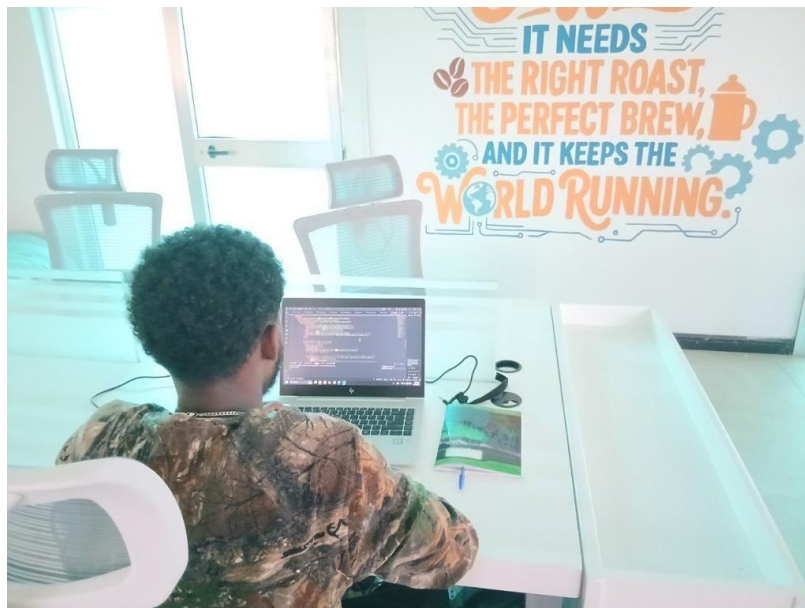


Figure 2.1: The author (Adisu Dereje) conducting full-stack web engineering at Qelem Meda Technologies PLC

2.4 Engineering Methods, Tools, and Techniques Used

Throughout the internship, I utilized modern software engineering methods and industry-standard tools:

- **Laravel 13 RESTful Architecture:** Employed PHP 8.3 and Laravel 13 as the backend foundation, utilizing Eloquent ORM for database relations, Form Request validation classes, API resource transformers, and database transactions.
- **Vue.js 3 & Modern Frontend Tooling:** Leveraged Vue 3 with the Composition API (<script setup>), Vite build tooling, Vue Router 4 for client-side routing, and Pinia for reactive global state management.
- **MySQL 8 Relational Database:** Structured a 3NF normalized schema in MySQL 8, applying foreign key constraints, unique indexes, and InnoDB row-level locking for atomic booking updates.
- **Git & GitHub Version Control:** Practiced Git feature-branch workflows, creating pull requests, resolving merge conflicts, and participating in peer code reviews before staging integration.
- **API Testing with Postman:** Utilized Postman for testing HTTP endpoints, verifying JSON schemas, testing authentication headers, and inspecting server response status codes.
- **UI/UX Design Principles:** Developed clean, user-friendly layouts prioritizing mobile responsiveness, clear visual hierarchy, accessible form elements, and immediate interactive feedback.

2.5 Major Challenges and Problems Faced

During my internship at Qelem Meda Technologies PLC, I encountered a variety of real-world challenges spanning daily transportation logistics, socio-economic factors, workplace infrastructure, and technical software engineering. Experiencing these challenges firsthand provided deep insight into the practical realities of professional life in urban Ethiopia:

2.5.1 Environmental, Logistical, and Commuting Challenges

- **1. Substantial Geographical Commute Distance:** I reside in the Nifas Silk-Lafto Sub-City, specifically in the Gofa Camp neighborhood, while the headquarters of

Qelem Meda Technologies PLC is situated around Sealite Mihret (in the Yeka / CMC corridor). The extreme cross-city transit distance between these two distant parts of Addis Ababa meant that commuting each day demanded substantial travel time, requiring extraordinary planning to arrive on schedule.

- **2. Severe Public Transportation Queues and Attendance Punctuality:** Navigating peak-hour public transportation was a major daily struggle. Overwhelming minibus taxi queues (የታክሲ ሰልፍ) both during the early morning rush and late evening return trips made traveling unpredictable. The pressure to beat these extensive queues to arrive on time and sign the official morning attendance sheet (as mandated by company policy) created persistent daily urgency and physical fatigue.
- **3. Late-Evening Neighborhood Security Checkpoint Restrictions:** Due to prolonged evening taxi delays across multiple transit terminals (such as Mexico, Stadium, or Megenagna), I often arrived back in Gofa Camp late at night. Because Gofa Camp is an active military and residential compound, strict evening security checkpoints required producing official Kebele residency identification (የካቢራ ግዛት መታወቂያ). On days when delays occurred or verification took time, navigating security clearance after an exhaustive workday proved challenging.
- **4. Financial Constraints on Multi-Leg Transit Fares:** As a self-supported undergraduate student intern, managing the escalating daily public transportation expenses across multi-leg minibus taxi transfers (frequently requiring two to three transit hops each way) posed tangible financial strain over the two-month duration.
- **5. Periodic Office Internet and Network Disruptions:** The development team occasionally encountered temporary internet service interruptions and bandwidth degradation at the office. These outages temporarily delayed downloading Composer packages, npm node_modules dependencies, and pushing code updates to remote GitHub repositories.

2.5.2 Technical and Workflow Engineering Challenges

- **1. Adaptation to Modern Production Tech Stack:** Transitioning from university classroom programming to a production-grade tech stack (PHP 8.3 with Laravel 13, Vue.js 3 with Composition API, and utility-first Tailwind CSS) initially presented a

steep learning curve. Mastering concepts like asynchronous Pinia store actions, Form Request lifecycle hooks, and Eloquent model scopes required intensive self-study.

- **2. Defect Tracking and Workflow Collaboration in Notion:** During quality assurance testing of commercial platforms—most notably the candidate registration system on eSchool (eSchool.qmt.et)—the engineering team utilized Notion databases for logging bug tickets, reproduction steps, and severity tags. As a newcomer to Notion's enterprise project management databases, categorizing defects properly and communicating edge-case test results clearly with senior QA leads initially posed a coordination challenge.
- **3. Database Concurrency and Tour Booking Collisions:** In the BetLink platform, managing concurrent booking requests where multiple prospective tenants attempt to reserve the identical tour inspection slot simultaneously posed critical race condition hazards requiring rigorous database-level isolation.

2.6 Measures Taken and Solutions Implemented

To overcome the practical and technical challenges I encountered during my internship, I adopted disciplined daily habits, practical problem-solving methods, and consistent communication with my supervisor and colleagues:

2.6.1 Logistical Adaptation and Professional Discipline

- **1. Early Departure and Transit Route Selection:** To consistently overcome the transit distance between Gofa Camp and Sealite Mihret and bypass peak minibuss queues, I committed to waking up early (departing before 6:30 AM). By selecting faster transfer routes and departing early, I successfully maintained a punctual record on the daily company attendance sheet.
- **2. Credential Management and Security Compliance:** To resolve security clearance hurdles at Gofa Camp checkpoints, I carried my verified BDU university identification, Kebele residency credentials, and official Qelem Meda internship letter, ensuring respectful and smooth access even during late-evening returns.
- **3. Offline Development and Local Environment Caching:** During workplace internet outages, I maintained steady progress by setting up local offline

development environments (local MySQL and local PHP artisan server), pre-caching essential npm and composer packages, and drafting component designs offline before syncing during network stability.

- **4. Intensive Self-Study and Mentorship Engagement:** I dedicated evening and weekend hours to studying the official documentation for Laravel 13, Vue.js 3, and Tailwind CSS. Under the patient technical mentorship of Mr. Samuel Temesgen, I rapidly bridged knowledge gaps and gained production-level confidence within the initial two weeks.
- **5. Standardized Bug Reporting in Notion:** I quickly familiarized myself with Notion's workspace templates by reviewing prior defect logs. I standardized my bug submissions with exact URL endpoints, HTTP payload dumps, expected vs. actual behaviors, and annotated screenshots, significantly improving bug-triage speed for the team.

2.6.2 Contributions and Value Delivered to the Hosting Organization

Throughout my two-month internship, my efforts translated into practical contributions to Qelem Meda Technologies PLC:

- **1. Practical Quality Assurance & Defect Reporting:** I participated in quality assurance testing and defect verification on the live eSchool candidate registration portal (eSchool.qmt.et) and the Haleta ERP system (haleta.qelemmeda.et). I tested form validations, country and tax center configurations, and modal dialogs, identifying edge-case errors, documenting them in Notion with reproduction steps and screenshots, and reporting them during team standups, as illustrated in Figure 2.2.
- **2. Practical Web Development on BetLink:** Under my supervisor's guidance, I contributed to the BetLink Property Management System by creating normalized database tables in MySQL, implementing backend CRUD API routes and controllers in Laravel 13, and developing responsive frontend components in Vue 3.
- **3. Team Collaboration and Task Execution:** By completing my assigned tasks steadily, following Git version control guidelines, and participating actively in team discussions, I learned to work cooperatively in a professional environment and helped handle routine development tasks.

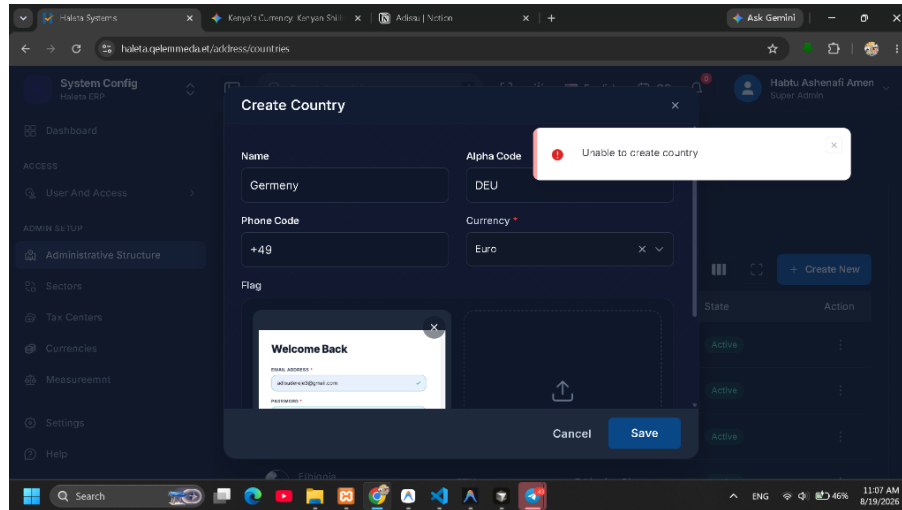


Figure 2.2: Quality assurance testing and error validation on Haleta ERP with Notion defect tracking (Adisu Dereje)

2.7 Personal Gains and Development

The two-month internship provided me with meaningful personal and academic growth across several key areas:

2.7.1 Practical Skills

During my two-month training, I gained practical programming skills. I learned how to structure backend PHP controllers in Laravel, build interactive Vue 3 components, write structured MySQL queries with joins, and test HTTP endpoints and error responses using Postman and browser developer tools.

2.7.2 Theoretical Knowledge

The internship helped reinforce what I learned at university. Concepts of object-oriented programming, relational database normalization, and RESTful architecture became clearer when I applied them to actual project features.

2.7.3 Teamwork and Communication

Working in a team environment taught me the importance of communication. Attending daily standup meetings, asking for guidance when stuck, and coordinating code changes with Git helped me understand how team members collaborate effectively.

2.7.4 Personal Responsibility and Task Ownership

I was assigned specific tasks including parts of the appointment scheduling workflow and QA bug logging. Taking responsibility for completing these tasks on time helped me develop personal discipline and practical problem-solving habits.

2.7.5 Work Ethics and Professional Discipline

Being in a professional office environment strengthened my work discipline. Overcoming long transit commutes, arriving on time, respecting organizational rules, and learning from constructive feedback helped me adapt to workplace expectations.

2.7.6 Understanding the Software Industry

Observing how Qelem Meda Technologies builds software systems for clients gave me a realistic understanding of the software development field in Ethiopia. It showed me how technology projects are planned and delivered to meet client needs.

2.8 Recommendations and Conclusion

The internship at Qelem Meda Technologies PLC has been an invaluable learning experience that bridged the gap between academic study and industry reality. To further enrich the internship program, I recommend providing incoming interns with early access to internal code standards and development container setups, and organizing bi-weekly cross-departmental architecture discussions.

Chapter Three: Project Work – BetLink Property Management System

3.1 Summary of the Project

BetLink is a modern, web-based Real Estate and Rental Property Management System developed to transform how residential and commercial properties are listed, discovered, inspected, and rented in Ethiopian cities. The Ethiopian real estate and rental market has historically relied on informal brokers (delalas), paper listings, fragmented phone calls, and unverified social media posts. This informal model creates serious challenges: prospective tenants face inflated prices, hidden fees, inaccurate property descriptions, and risks of fraudulent transactions. Landlords struggle with high broker commissions, unreliable tenant screening, and uncoordinated property viewings.

BetLink provides a centralized, transparent digital ecosystem connecting property owners, prospective tenants, buyers, and verified agents. The platform delivers verified property listings, high-resolution multimedia galleries, dynamic multi-criteria search and filtering, automated online tour scheduling with conflict prevention, and direct landlord-tenant communication channels.

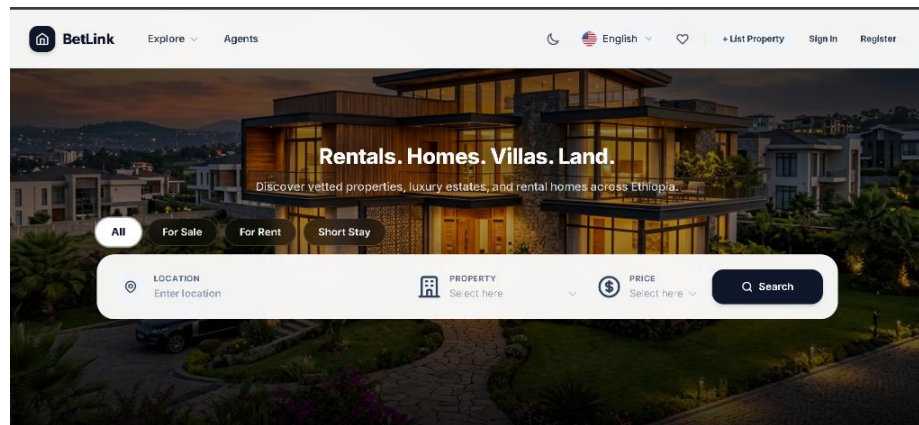


Figure 3.1: The BetLink Real Estate and Rental Property Management System Hero Interface

Key architectural layers and technologies of BetLink include:

- **Backend Layer:** Laravel 13 RESTful API backend handling business logic, database migrations, Eloquent ORM data relationships, Form Request validations, and JWT-authenticated session tokens.

- **Frontend Layer:** Vue.js 3 Single Page Application with Vite build system, Vue Router 4 for client routing, and Pinia for reactive global state management.
- **Database Layer:** MySQL 8 relational database organized in Third Normal Form (3NF) with foreign key constraints, composite indexes, and row-level locking.
- **Security Layer:** Two-factor authentication via SMS/Email OTP, password hashing using Argon2id/Bcrypt, role-based middleware, and CSRF/CORS protection.

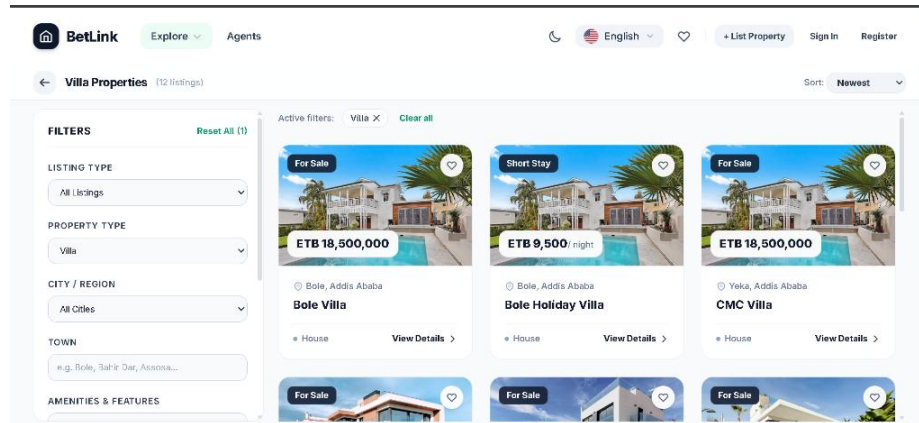


Figure 3.2: BetLink Verified Property Catalog and Filtering Interface

3.2 Problem Statement and Justification

3.2.1 Problem Statement

In urban Ethiopia, finding a rental home or commercial property is an inefficient, stressful, and expensive ordeal. The rental ecosystem suffers from acute market failures:

- **1. Informal Broker Exploitation:** Informal brokers charge steep, non-standard commission fees from both tenants and landlords while providing minimal accountability.
- **2. Fraudulent Listings:** Renters frequently encounter duplicate listings, misleading photos, non-existent properties, and deceptive pricing on social media channels.
- **3. Pricing Opacity:** Lack of centralized rental data leaves tenants without reliable knowledge of prevailing fair market rental rates across neighborhoods.
- **4. Disorganized Viewings:** Scheduling in-person viewings involves endless phone calls, missed meetings, and conflicting schedules with no centralized calendar.

3.2.2 Justification

BetLink was developed to systematically replace this broken manual system with an accessible, transparent, and reliable digital platform:

- **1. Cost Reduction:** By directly connecting landlords and tenants, BetLink eliminates predatory brokerage fees and promotes fair pricing.
- **2. Verified Transparency:** Every property listing requires landlord identity verification and title documentation before publication, eliminating fraud.
- **3. Convenience & Time Savings:** Users can browse high-resolution photos, verified amenity lists, and transparent rental terms from their phone or computer before visiting.
- **4. Tour Scheduling Convenience:** The integrated calendar lets tenants select available inspection time slots, preventing double-bookings and unnecessary travel.

3.3 Objectives of the Project

The primary objective of the BetLink project was to design, develop, and test an enterprise-grade property management web platform:

- **1. User-Friendly Platform:** Build a responsive Single Page Application that allows users to easily browse, filter, and view property details on any device.
- **2. Tour Scheduling Engine:** Implement an automated tour booking engine with concurrency locking to prevent overlapping appointments.
- **3. Security & Trust:** Enforce landlord verification workflows, two-factor OTP login, and strict role-based access control.
- **4. Landlord Dashboard:** Provide property owners with an intuitive dashboard to manage listings, track viewings, and respond to tenant inquiries.

Table 1.2: Objectives of the BetLink Property Management System

Objective	Description	Expected Outcome
User-Friendly Web Interface	Develop a modern responsive SPA using Vue 3 and Vite with dynamic search filters.	Intuitive property discovery across mobile and desktop devices.
Automated Tour Booking	Implement an appointment system with database transaction locking.	Zero duplicate bookings and synchronized landlord/tenant schedules.
Verified Listings & Trust	Enforce landlord documentation checks and	Elimination of ghost

	two-factor authentication.	listings, fraud, and misleading property claims.
Landlord Management Portal	Provide an analytics dashboard for property owners to manage listings and inquiries.	Streamlined property management and faster tenant onboarding.
RESTful API Architecture	Build scalable Laravel 13 endpoints with Eloquent ORM and Form Request validation.	Secure, maintainable, and high-performance backend infrastructure.

3.4 Methodology

The project followed an Agile Scrum development methodology organized into two-week sprints. This iterative approach allowed the team to incorporate continuous feedback from mentors and supervisors throughout the development lifecycle.

3.4.1 Requirement Analysis

The team conducted requirement gathering discussions with property owners, prospective tenants, and real estate professionals. The gathered insights were formalized into a Software Requirements Specification (SRS) defining functional requirements (user registration, listing submission, search filters, tour booking, messaging) and non-functional requirements (fast page load times, reliable system availability, secure password hashing, and responsive viewport scaling).

3.4.2 System Design

The system design phase translated requirements into detailed technical blueprints: a 3NF relational database schema in MySQL, UML sequence diagrams mapping the tour booking lifecycle, and component hierarchy wireframes for the Vue 3 frontend.

3.4.3 Development

Implementation proceeded across specialized frontend and backend tracks. Backend endpoints were developed in Laravel 13, tested via Postman, and integrated with the Vue 3 frontend using Axios client instances and Pinia stores. Daily standups ensured active collaboration between team members.

3.5 Analysis, Results, and Discussion

3.5.1 System Analysis

Before BetLink, prospective tenants spent an average of 2 to 3 weeks searching for suitable housing, visiting multiple properties that did not match phone descriptions, and paying unrecoverable broker registration fees. BetLink cuts this discovery cycle down to hours, enabling users to narrow down choices to verified properties matching their exact budget and amenities before scheduling a visit.

3.5.2 Results of the System

The completed BetLink platform successfully delivers all target capabilities: fast property searches with instant filtering, secure user authentication with OTP verification, dynamic property detail pages with interactive image galleries, and an automated appointment booking modal.

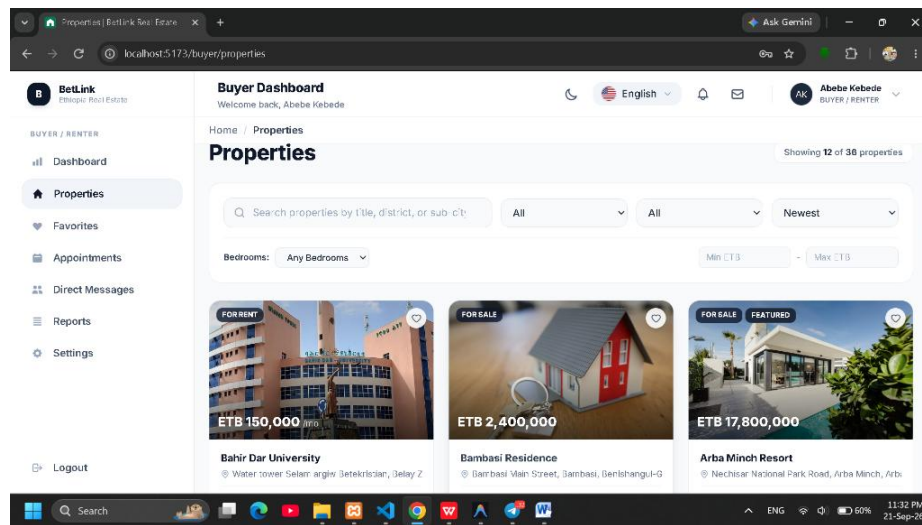


Figure 3.3: BetLink Buyer & Tenant Dashboard

3.5.3 Discussion

The practical development of BetLink demonstrates how modern web applications can directly address common everyday challenges in Ethiopia. By replacing unreliable manual intermediaries with transparent digital workflows, the platform provides clear economic value to both renters and property owners.

3.6 Proposed Solution

The proposed BetLink solution comprises four integrated pillars:

- **3.6.1 Digital Property & Tenancy Management:** Centralized digital registry for properties with verified ownership, structured pricing, and detailed amenities.
- **3.6.2 Real-Time Tour Scheduling & Updates:** Real-time tour scheduling engine with pessimistic database locking to guarantee collision-free appointments.
- **3.6.3 Transparency and Security:** Argon2id password encryption, two-factor OTP verification, role-based authorization, and transparent reviews.
- **3.6.4 Advantages of the Solution:** Direct landlord-tenant coordination, zero broker exploitation, verified listings, and cross-device accessibility.

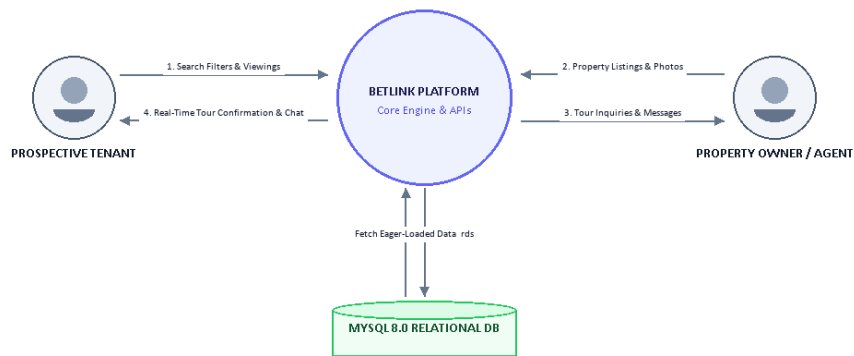


Figure 3.4: Data Flow Diagram for Tour Appointment Scheduling

Chapter Four: General Conclusion and Recommendation

4.1 General Conclusion

My two-month internship at Qelem Meda Technologies PLC has been a productive and educational experience. As an Information Technology student at Bahir Dar Institute of Technology, this internship allowed me to connect classroom lessons with practical tasks in an active software company.

Working on the BetLink platform gave me good practical exposure to web development using Laravel 13, Vue 3, Pinia, and MySQL. Under the guidance of my supervisor, I learned how backend routes and controllers operate, how to organize database tables, and how to test web application features.

Key learning outcomes of the internship include:

- **1. Application of Theoretical Knowledge:** Applied programming and database concepts learned in university to practical web development tasks.
- **2. Development of Practical Skills:** Learned practical fundamentals of Laravel 13, Vue 3, and MySQL database operations.
- **3. Enhancement of Teamwork Skills:** Practiced asking constructive questions, participating in daily team standups, and coordinating work through Git.
- **4. Understanding Work Ethics:** Learned the importance of punctuality, professional discipline, and task responsibility in a workplace.
- **5. Industry Awareness:** Observed how local software companies develop practical digital solutions for clients in Ethiopia.

The mentorship of Mr. Samuel Temesgen and academic guidance from Mr. Asnakew Lingerih gave me valuable practical advice and helped me build steady confidence for my future academic and professional development.

4.2 General Recommendations

Based on my internship experience, I propose the following recommendations:

- **1. For Bahir Dar Institute of Technology (BiT):** Incorporate modern web frameworks (such as Laravel and Vue.js) and collaborative Git workflows into undergraduate coursework.

- **2. For Qelem Meda Technologies PLC:** Provide incoming interns with structured onboarding sessions covering internal coding standards and containerized environments.
- **3. For Platform Expansion:** Integrate local mobile payment gateways (such as Telebirr and Chapa) into BetLink to support digital booking deposits.
- **4. For Future Interns:** Proactively learn modern development tools, engage actively during team discussions, and maintain consistent engineering notes.

4.3 Final Remarks

Overall, the internship at Qelem Meda Technologies PLC has laid a solid foundation for my professional journey in software engineering. Working on the BetLink platform gave me practical exposure to how modern digital technologies can solve real-world challenges, improve efficiency, and enhance user satisfaction. The lessons learned will guide me as I strive to become a competent and responsible IT professional.

References

1. Qelem Meda Technologies PLC Official Corporate Profile and Documentation – Addis Ababa, Ethiopia.
2. Internal Company Guidelines and Quality Policies – Qelem Meda Technologies PLC.
3. Laravel 13.x Documentation – <https://laravel.com/docs/11.x>
4. Vue.js 3 Documentation – <https://vuejs.org/guide/introduction.html>
5. MySQL 8.0 Reference Manual – <https://dev.mysql.com/doc/refman/8.0/en/>
6. Sommerville, Ian. Software Engineering, 10th Edition. Pearson, 2015.
7. Pressman, Roger S. Software Engineering: A Practitioner's Approach, 8th Edition. McGraw-Hill, 2011.
8. International Organization for Standardization. ISO/IEC 27001:2022 Information Security Management.
9. International Organization for Standardization. ISO 9001:2015 Quality Management Systems.

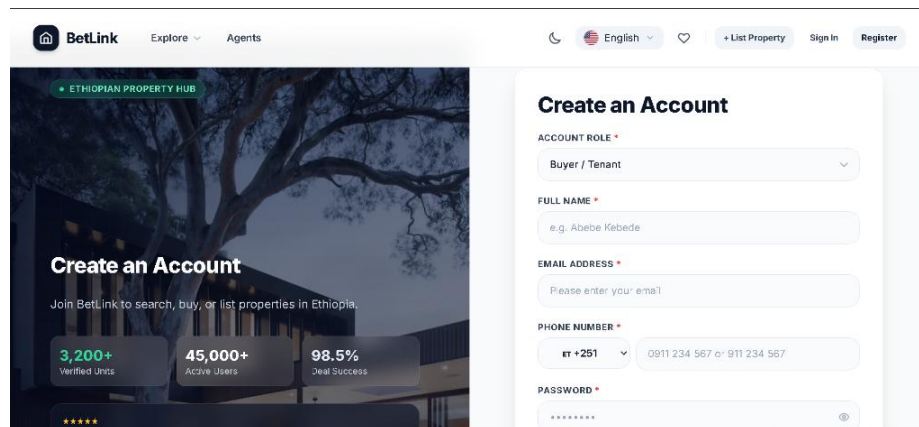
Appendices

The appendices provide supplementary information, supporting materials, and documentation relevant to the internship experience and the development of the BetLink Real Estate and Rental Property Management System.

Appendix A: Sample Screenshots of the BetLink Platform

This appendix presents visual documentation of the platform's key features and user interfaces. The screenshots capture the functionality, design, and usability of the application:

- **User Login & Registration Screens:** Demonstrating secure access and two-factor OTP verification.
- **Property Catalog & Search Interface:** Displaying dynamic search filters, property cards, and location-based results.
- **Property Details & Tour Scheduling:** Showing full specifications, image carousels, verified amenities, and tour booking modal.
- **Landlord Management Dashboard:** Illustrating listing management, appointment status tracking, and inquiry analytics.



Appendix A.1: BetLink User Registration and Onboarding Screen

Appendix B: Code Snippets and Features Implemented

This appendix highlights key code segments and modules implemented during the internship, including user registration, appointment controllers, concurrency locking, and state management:

- **User Registration & Authentication:** Laravel 13 API authentication controller showing user registration logic and token generation within the active development workspace (Adisu@HP).
- **Tour Booking Controller:** Controller methods enforcing active property checks, duplicate booking prevention, and ownership validation.
- **Concurrency Protection:** Pessimistic database locking ensuring time slots cannot be simultaneously booked by two prospective tenants.

```

1 namespace App\Http\Controllers\Api\V1\Auth;
2
3 use App\Http\Controllers\Controller;
4 use App\Http\Requests\Auth\RegisterRequest;
5 use App\Models\User;
6 use App\Traits\ApiResponse;
7 use Illuminate\Http\JsonResponse;
8 use Illuminate\Support\Facades\Hash;
9
10 class RegisterController extends Controller
11 {
12     use ApiResponse;
13
14     public function register(RegisterRequest $request): JsonResponse
15     {
16         $role = match(strtolower($request->role ?? 'buyer')) {
17             'buyer', 'buyer / tenant', 'buyer-tenant', 'buyer / renter', 'buyer-renter' => 'buyer',
18             'owner', 'property owner', 'property owner' => 'owner',
19             'agent', 'real estate agent', 'real estate agent' => 'agent',
20             default => 'buyer';
21         };
22
23         $user = User::create([
24             'name' => $request->name ?? trim($request->first_name ?? '' . ' ' . ($request->last_name ?? '')),
25             'email' => $request->email,
26             'password' => Hash::make($request->password),
27             'role' => $role,
28         ]);
29
30         $token = $user->createToken('api-token');
31
32         return $this->response([
33             'user' => $user->toArray(),
34             'token' => $token->plainTextToken,
35         ], 201);
36     }
37 }

```

Appendix B.1: Implementation of User Registration API in Laravel 13 with Active Development Terminals (Adisu@HP)

```

1 namespace App\Http\Controllers\Api\V1\Appointment;
2
3 use App\Http\Controllers\Controller;
4 use App\Http\Requests\Appointment\StoreAppointmentRequest;
5 use App\Models\Appointment;
6 use App\Models\Property;
7 use App\Traits\ApiResponse;
8 use Illuminate\Http\JsonResponse;
9 use Illuminate\Support\Facades\Hash;
10
11 class AppointmentController extends Controller
12 {
13     use ApiResponse;
14
15     public function store(StoreAppointmentRequest $request): JsonResponse
16     {
17         $property = Property::where('status', 'active')->findOrFail($request->property_id);
18
19         if ($property->user_id === $request->user->id) {
20             return $this->error('You cannot book an appointment for your own property', 422);
21         }
22
23         // Prevent duplicate active viewing appointments for the same property
24         $existing = Appointment::where('property_id', $property->id)
25             ->where('visitor_id', $request->user->id)
26             ->where('status', ['pending', 'confirmed'])
27             ->first();
28
29         if ($existing) {
30             $date = $existing->scheduled_at ? $existing->scheduled_at->format('M d, Y \at h:i A') : 'a scheduled time';
31             return $this->error("You already have an active ([{$existing->status}]) viewing appointment for this property on [{$date}]. You can reschedule or cancel it.", 422);
32         }
33
34         $appointment = $this->appointmentService->book($property, $request->user(), $request->validated());
35         return $this->response($appointment, 'Appointment booked successfully', 201);
36     }
37
38     /** @return JsonResponse */
39     public function show(Appointment $appointment): JsonResponse
40     {
41         $this->authorize('view', $appointment);
42         return $this->success($appointment->load(['property.primary_image', 'property.address.city', 'property.address.subcity', 'owner', 'visitor']));
43     }
44
45     /** @return JsonResponse */
46     public function confirm(Appointment $appointment, Request $request): JsonResponse
47     {
48         $this->authorize('manage', $appointment);
49         $appointment->confirm($request->confirmed_at);
50         return $this->response($appointment, 'Appointment confirmed', 200);
51     }
52 }

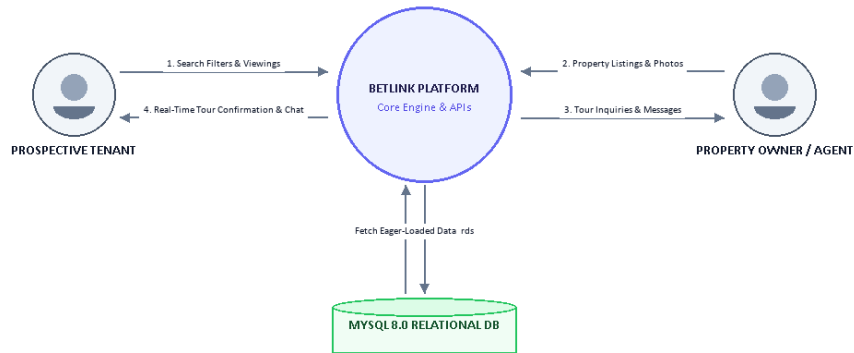
```

Appendix B.2: AppointmentController.php - Tour Booking and Concurrency Locking

Appendix C: Flowcharts and System Architecture Diagrams

This appendix provides visual representations of system architecture, database data flow, and user interaction cycles:

- **Data Flow Diagram for Tour Booking:** Illustrating request-response cycles between the client, API layer, and MySQL database.
- **Database Entity Relationship Diagram (ERD):** Mapping relational entities including users, properties, appointments, and reviews.



Appendix C.1: BetLink Architecture and Data Flow Overview